Calcurion Maths Worksheets

GCSE Mathematics

Topic: Graphs

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find the gradient of the line joining $(1, 2)$ and $(3, 6)$. **[2]**

2. Find the gradient of the line joining $(0, 4)$ and $(2, 0)$. **[2]**

3. Find the equation of the line with gradient 3 that passes through $(0, 2)$. **[2]**

4. Find the equation of the line with gradient -2 that passes through $(0, 5)$. **[2]**

5. Find the y-intercept of the line $y = 4x - 7$. **[1]**

6. Find the gradient of the line $y = 5 - 3x$. **[1]**

7. Does the point $(2, 7)$ lie on the line $y = 3x + 1$? **[2]**

8. Find the midpoint of the line segment joining $(2, 4)$ and $(6, 10)$. **[3]**

9. Find the distance between the points $(1, 2)$ and $(4, 6)$. **[3]**

10. Find the equation of the line parallel to $y = 2x + 3$ that passes through $(0, 5)$. **[2]**

11. Find the equation of the line perpendicular to $y = 3x - 1$ that passes through $(0, 4)$. **[3]**

12. Sketch the graph of $y = 2x + 1$, showing the intercepts. **[3]**

13. Sketch the graph of $y = x^2$, showing the vertex. **[3]**

14. Find the coordinates of the turning point of $y = x^2 - 4x + 3$. **[3]**

15. Find the roots of $y = x^2 - 5x + 6$. **[3]**

Section B – Hard (Q16–25)

16. Find the equation of the line that passes through $(1, 3)$ and $(4, 9)$. **[4]**

17. Find the equation of the line that passes through $(2, 5)$ and is parallel to $y = 4x - 1$. **[3]**

18. Find the equation of the line that passes through $(3, 2)$ and is perpendicular to $y = \frac{1}{2}x + 1$. **[4]**

19. The line $y = mx + c$ passes through $(2, 7)$ and has gradient 3. Find c. **[3]**

20. Sketch $y = (x - 2)(x + 1)$, showing intercepts and turning point. **[5]**

21. Find the intersection of the lines $y = 2x + 1$ and $y = x + 4$. **[3]**

22. Find the intersection of $y = x^2$ and $y = 2x + 3$. **[5]**

23. A straight line has equation $3x + 2y = 12$. Find the gradient and y-intercept. **[3]**

24. Find the equation of the circle with centre $(0, 0)$ and radius 5. **[2]**

25. Does the point $(3, 4)$ lie on the circle $x^2 + y^2 = 25$? **[2]**

End of Worksheet • *Show all working clearly.*